Graphic Era HILL UNIVERSITY
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End Term (Even) Semester Regular/Back Examination May 2026

Roll no...2594075.............

Name of the Program and semester: **B. Tech (AIML and AIDS)**
Name of the Course : **Fundamentals of Artificial Intelligence and Machine Learning**
Course Code: **TCS 203**
Time: 3 hours Maximum Marks: 100

Note:
(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

List and explain the different types of intelligent agents with examples: **CO1**

i. Simple Reflex Agent
ii. Model-Based Agent
iii. Goal-Based Agent
iv. Utility-Based Agent

b. A bank uses Logistic Regression to predict whether a loan applicant will default (1) or not default (0) based on two features: **CO3**

- x1 = Income Score
- x2 = Credit Score

The model parameters are: w1 = 0.6, w2 = -0.4, b= 0.3. For a point X = (3,2) compute the following:

i. Compute the linear combination z = w1x1+w2x2.
ii. Apply sigmoid function to compute the predicted probability.
iii. Predict the class label using threshold = 0.5.
iv. If actual class is 1, compute the Loss.

c. Describe the following terms with the help of an example: **CO1**

i. Actuators
ii. Environment
iii. State
iv. Agent
v. Performance

Q2. (2X10=20 Marks)

a. Explain K-Means Clustering and Hierarchical Clustering in brief. Compare both techniques based on working principle, advantages, and limitations. Also evaluate which clustering method is more appropriate for large datasets. **CO5**

b. Describe the working of the Apriori Algorithm for association rule learning with the help of a given example. Explain how frequent itemsets are generated using minimum support and how association rules are formed using minimum confidence. **CO6**

Transaction_ID	Items_Purchased
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter

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c. A manufacturing company studies the relationship between Machine Operating Hours and Units Produced. Given the dataset having machine operating hours as feature and production output as target. **CO3**

Machine Operating Hours(X)	1	3	5	7	9
Units Produced(Y)	3	6	7	10	12

i. Compute the slope and intercept of the regression line.
ii. Find the equation of best fit line.
iii. Predict the Units Produced when the machine operates for 9 hours.
iv. Calculate the MAE (Mean Absolute Error) and MSE (Mean Squared Error) for the dataset.

Q3. (2X10=20 Marks)

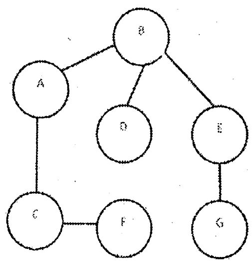
a. A hospital has deployed an AI model to detect whether a patient has a **Serious Disease (Class 1 = Disease, Class 0 = Healthy)**. Below is the record of 10 patients evaluated during a pilot test: **CO4**

Actual = [1,0,1,1,0,0,1,0,1,0]
Predicted = [1,0,1,0,0,1,1,0,1,0]

i. Construct the confusion matrix for the data provided.
ii. Calculate the accuracy of the model.
iii. Compute the Precision of the model.
iv. Find the Recall of the model.
v. Compute the F1-Score of the model.

b. Explain the working of A* algorithm with the help of an example. **CO2**

c. Consider the following graph with starting node as A and goal node as G. Apply DFS algorithm to find the path from starting node to goal node and display the use of stack for each node. **CO2**



Q4. (2X10=20 Marks)

a. Describe the following terms with respect to SVM and demonstrate using a diagram. **CO3**
   i. Kernels
   ii. Hyperplanes
   iii. Margin
   iv. Support Vector

b. A hospital wants to classify whether a patient is **High Risk (H)** or **Low Risk (L)** based on two health indicators: X1 = Blood Pressure Score, X2 = Cholesterol Score. Apply KNN with K= 3 to

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classify the patient with feature X = (6, 6) **CO3**

Patient	P1	P2	P3	P4	P5	P6	P7	P8
X1	2	3	4	6	7	8	5	9
X2	3	2	4	5	7	6	3	8
Risk Class	L	L	L	H	H	H	L	H

c. Explain the need for **regularization** in regression models. Describe the working of **Ridge Regression** and **Lasso Regression** with their cost functions, and compare both techniques in terms of coefficient shrinkage, feature selection, and applications. **CO3**

Q5. (2X10=20 Marks)

a. A retail company wants to segment its customers based on their **monthly purchase amount** (in ₹1000) and **Number_of_Visits_Per_Month**. The management decides to group customers with similar spending patterns. The monthly purchase amounts of 8 customers are given below:

Customer	C1	C2	C3	C4	C5	C6	C7	C8
Amount	1	2	5	6	10	11	15	16
Visits	2	3	4	5	8	9	10	11

Apply K-means clustering with k=2. **CO5**

b. Develop a code to implement **Logistic Regression** on a dataset containing features **StudyHours, Attendance, PreviousMarks, AssignmentScore** and **PassStatus** as label. The code should perform the following: **CO6**

i. Load dataset from a csv file named student_dataset.csv.
ii. Extract the features and labels.
iii. Split the dataset into training and testing sets in the ratio of 80:20.
iv. Train the logistic regression model on the training data and make predictions on the testing data.
v. Display accuracy, precision, recall, F1-score, and confusion matrix of the model.

c. i. Demonstrate the working of Decision Tree Classifier discussing the importance of Gini Index. **CO4**
ii. Consider the following dataset for predicting whether a student will **Pass** or **Fail** based on **Study_Hours** and **Attendance using Decision Tree Classifier**.

Student	Study_Hours	Attendance	Result
S1	High	High	Pass
S2	High	Low	Pass
S3	Low	High	Fail
S4	Low	Low	Fail